

NETWORKED CONTROL

26/06/2023

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EXERCIZES

1) Consider the following continuous-time model of system $\mathcal{S}$.

$$\dot{x}(t) = -x(t) + 2u(t)$$

$$y(t) = 2x(t)$$

A. Answer to the following questions.

- Compute the eigenvalue and the mode of the model. Is the system open-loop asymptotically stable?
- Assume that the system is controlled using a continuous-time control law of the type $u(t) = Ky(t)$. Define the range of values of K that make the closed-loop system asymptotically stable.

B. Compute the parameters F, G , and H of the discrete-time dynamical model

$$x_{k+1} = Fx_k + Gu_k, \quad y_k = Hx_k$$

of system $\mathcal{S}$ with sampling time $h = \log_e 2$ s.

Also, assuming that the system is controlled using a digital control law of the type $u_k = Ky_k$, define the range of values of K that make the closed-loop system asymptotically stable.

C. Assume that the system $\mathcal{S}$ is controlled through the networked control system (NCS) sketched in the following figure.

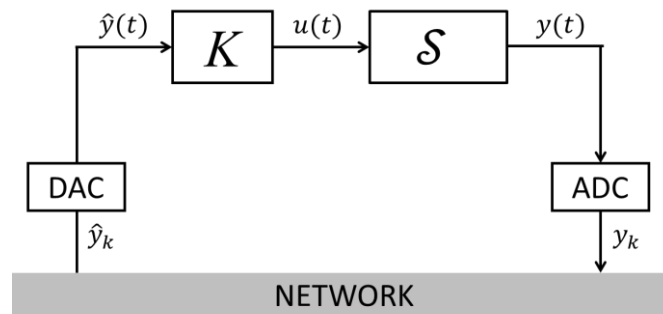


Figure 1: Networked control system

- Assume that $h = \log_e 2$ s and $K = -\frac{1}{2}$ and that there are no dropouts. What is the maximum value $\bar{\tau}$ of the delay τ such that the NCS is asymptotically stable iff $\tau < \bar{\tau}$?
- Assume now that the NCS is subject to packet dropouts, that the delay is $\tau = \log_e 1.5$ s, and that $h = \log_e 2$ and $K = -\frac{1}{2}$. Variable θ_k has the following periodic behavior.

k	1	2	3	4	5	6	7	8	9	...
θ_k	1	0	0	1	0	0	1	0	0	...

Is the NCS asymptotically stable?

2) Shortly discuss how the **choice of the sampling time** is carried out in **digital control systems**. Explain how, qualitatively, the choice of the **sampling time** affects the **performance of a networked control system**, in case the control network is affected by delays and possible packet dropouts.

3) Assume that the linear system

$$\begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) \end{cases}$$

is controlled through the **networked control system (NCS)** sketched in the Figure 1.

Assume that the **sampling time** h_k and the **delay** τ_k affecting the control network are **time varying** and that $h_{min} \leq h_k \leq h_{max}$, $\tau_{min} \leq \tau_k \leq \tau_{max}$. State the main theorem providing the **condition guaranteeing the exponential stability of the NCS**. Is the condition sufficient, necessary, or both? Why?

Also, comment of the **numerical solvability of the condition** provided in the theorem and on the **conservativeness of the result**.